



Islamic University of Technology

Department of Computer Science and Engineering (CSE)

CSE 4206: Digital Logic Design Lab
Summer, 2024-2025

Faisal Hussain, Assistant Professor, CSE
Aashnan Rahman, Jr. Lecturer, CSE

Lab 4

Getting Started With Simulation Software Proteus

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1 Introduction

In the previous labs, we implemented various Boolean circuits using a trainer kit. From now on, all tasks will be performed individually using Proteus, a simulation software. A classroom assignment will be posted at the beginning of each lab session. You will have about 1 hour to complete the task and submit it with the proper file naming format. Each task will be evaluated based on a specific rubric.

Warning: If you are caught copying another classmate's lab task, you will receive a zero for that specific lab.

2 Proteus

Proteus is an electronic design automation (EDA) software developed by Labcenter Electronics. It combines circuit simulation, schematic capture, and PCB layout in a single platform. Widely used in education and industry, Proteus is especially known for simulating microcontroller-based designs, allowing users to test both hardware and firmware before building real prototypes.

3 Getting Started with Proteus

3.1 Install Proteus

You can install proteus from the [provided drive link](#) or any other source.

3.2 Create a File

1. Open Proteus.
2. Go to **New Project**, choose file name and path and click next.
3. Choose "Create a schematic from the selected template" and then choose "Default".
4. Then choose "Do not create a PCB layout" and click next.
5. Choose "No firmware project" and proceed.
6. Click "Finish".
7. If you get something similar to [Figure 1](#) then you have been able to create a file properly.

3.3 Adding Basic Components to your Workspace

1. Click on the **P** button on the left bar (next to devices).
2. Add the basic Input and Output components: LOGICTOGGLE for input and LOGICPROBE(BIG) for output. You can search for these components using keywords and add them by double-clicking on the desired item in the results on the right.

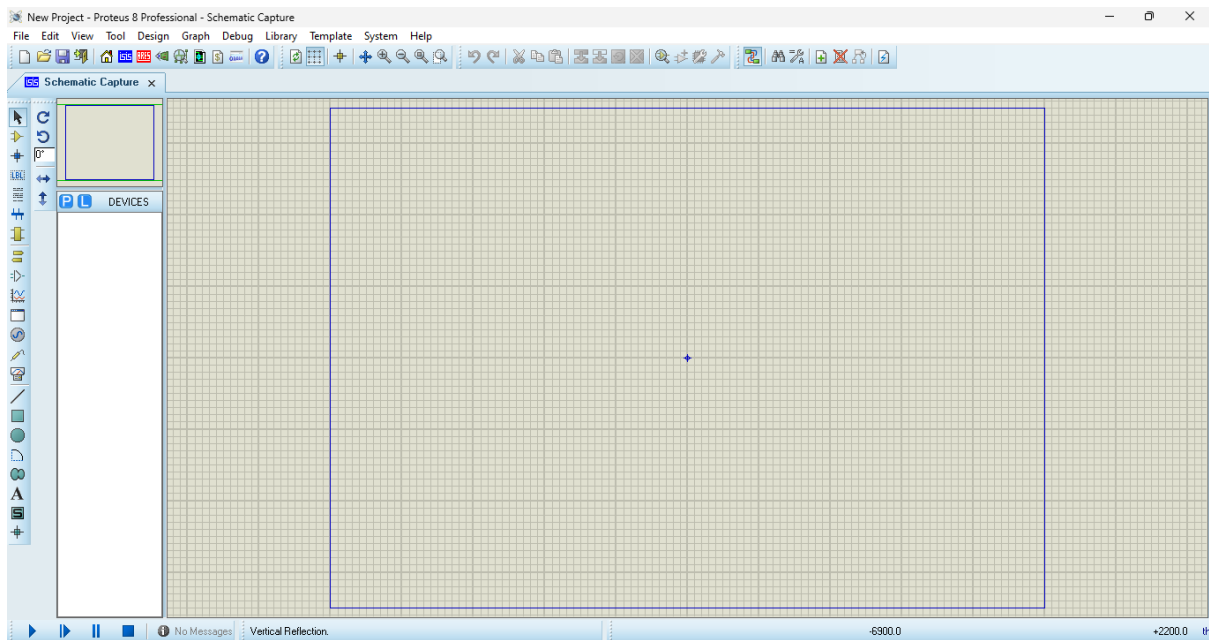


Figure 1: Empty Proteus File

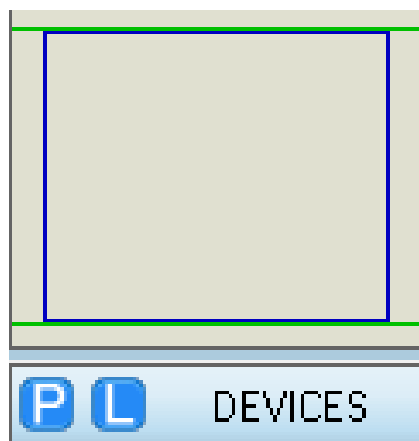


Figure 2: Pick Devices

3. To check a sample circuit, add another component, for example a X-OR gate in the similar way and build the following circuit.

To place the component in your workspace, choose a device from the DEVICES panel and click in your workplace. Connect the components using wire.

4. Press on the simulation icon and try out different combinations.

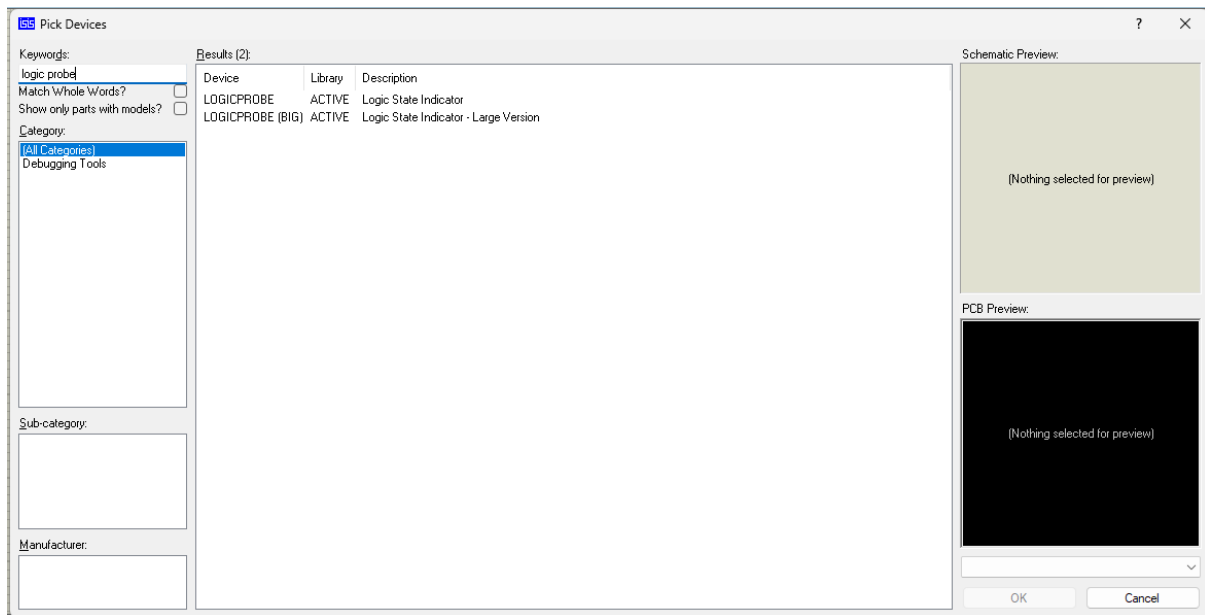


Figure 3: Adding Components

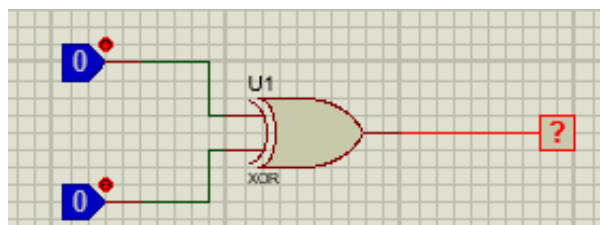


Figure 4: Basic X-OR Circuit

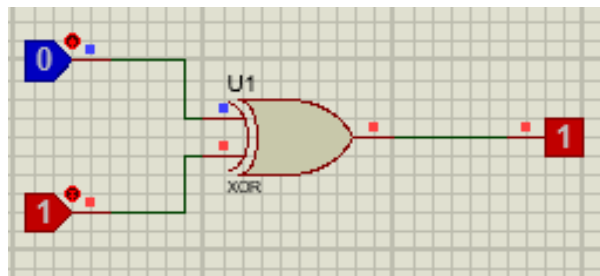


Figure 5: Simulation

4 Binary Subtractor

A binary subtractor is a digital logic circuit that performs subtraction of binary numbers. Just like an adder adds binary values, a subtractor computes the difference between two binary digits.

There are two main types:

1. **Half Subtractor:**

Subtracts two single bits A and B.

2. **Full Subtractor:**

Subtracts three bits: A, B and Borrow-in (B_{in}) from a previous stage.

5 Experiment 4: Implementation of Subtractor Circuits

5.1 Objectives

The objectives of this experiment are as follows:

1. To understand and implement binary Half Subtractor and Full Subtractor circuits.
2. To design and implement a 4-bit binary adder.

5.2 Tasks

Today's lab task are as follows -

1. Implement a Half Subtractor circuit using:
 - (a) Only basic gates (AND, OR, NOT)
 - (b) Any gates (simplified form, e.g., XOR)
2. Implement a Full Subtractor circuit using two Half Subtractor circuits and additional logic gates.
3. Implement a 4-bit circuit that subtracts two numbers using radix complement (2's complement method). The circuit should handle both positive and negative results appropriately.

6 Instructions:

1. **File Naming Format:** ID_L04T01
2. You will be provided with an A4 sheet on which you must **write your ID at the top**. For Task 1 and Task 2, include the following:
 - (a) Truth Table
 - (b) Boolean expressions of the simplified circuit
 - (c) Simplification (if required) using K-map or cut and try method.
3. For Task 3, you must show the thought process behind the circuit design, providing appropriate justification.
4. You will get 1 hour to complete these tasks.
5. The rubrics for today's lab task can be found [here](#).